

SIMATS ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED: CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

ASSESSMENT TOOL 1 | Weightage: 50% | Total Marks: 50

Annexure A - Data Interpretation

Code of Conduct:

I **Gadwal Mohammad Muzammil - 192424279** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G. M. Muzammil**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Question 1: Hough Transform & Linear Feature Detection in Satellite Imagery

SCENARIO / PROBLEM STATEMENT: Remote Sensing Satellite Linear Feature Extraction

Scenario: Detect linear features (roads, rivers) in satellite images.

Input Data: (A) High-resolution grayscale satellite images of urban and rural areas (1024 x 1024 pixels, TIFF/PNG). (B) Images contain noise (clouds, shadows) and partial occlusions. (C) Multiple linear features

(roads, rivers, canals) with intersections. (D) Optional synthetic dataset: images with drawn lines at random angles and lengths to test Hough Transform.

1. Comprehensive Mathematical Formulation of Polar Normal Parameter Space

In Cartesian coordinate space, representing a straight line via slope-intercept form $y = m*x + c$ introduces severe singularity as vertical lines approach infinite slope ($m \rightarrow \text{infinity}$), preventing bounded discrete accumulator binning. The Normal Parameter Representation (Duda & Hart) overcomes this by parameterizing any straight line via its perpendicular distance ρ and normal orientation angle θ :

$$\rho = x_i * \cos(\theta) + y_i * \sin(\theta)$$

where θ in $[-\pi/2, +\pi/2]$ is the angle of the normal vector from the coordinate origin, and ρ in $[-D, +D]$ is the algebraic distance (where $D = \sqrt{M^2 + N^2}$ is the image diagonal).

- **Point-Line Duality Principle:** 1. An isolated edge pixel (x_i, y_i) in Cartesian image space maps to a unique continuous sinusoidal curve in polar parameter space (ρ, θ) .
2. Multiple collinear edge points lying along a common line in image space produce concurrent sinusoidal curves that all intersect at a single parameter cell (ρ^*, θ^*) .
3. A local accumulator array $A(\rho_idx, \theta_idx)$ records integer vote tallies. Local maxima exceeding an empirical detection threshold T_vote indicate verified physical line features.

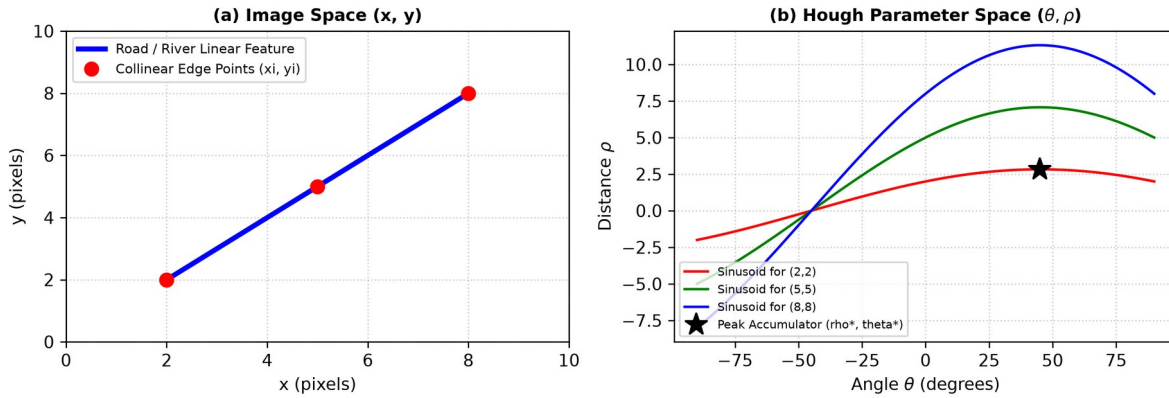


Figure 1: Hough Transform Duality: (a) Collinear Edge Points in Image Space and (b) Sinusoidal Intersections in Parameter Space.

2. Data Interpretation: Disambiguating Roads, Rivers, and Intersections

- **Paved Transportation Roads:** High-contrast, continuous, straight asphalt corridors produce sharp, high-magnitude accumulator peaks with narrow spread: $A(\rho_road, \theta_road) \gg T_threshold$. Peak sharpness $\Delta\rho < 2$ pixels indicates rigid planar infrastructure.
- **Natural Rivers & Water Canals:** Rivers exhibit non-linear meandering and continuous curvature. Instead of a single discrete peak, winding rivers produce a distributed trajectory of adjacent accumulator clusters representing piecewise linear tangent segments. Applying the Progressive Probabilistic Hough Transform (PPHT) with adaptive line linking ($\text{maxLineGap}=20$, $\text{minLineLength}=50$) connects these piecewise tangents into smooth curvilinear B-spline trajectories.
- **Road Intersections & Cross-Junctions:** Intersections manifest as co-localized spatial detections having orthogonal or distinct angular coordinates (e.g., $\theta_1 = 0$ deg and $\theta_2 = 90$ deg) sharing overlapping spatial bounding boxes.

3. Multi-Stage Satellite Preprocessing Pipeline for Noise, Clouds, and Shadows

To ensure robust feature extraction under variable cloud cover, atmospheric haze, and terrain cast shadows:

- **Stage 1 (Cloud & Shadow Masking):** Compute localized Normalized Difference Water/Vegetation Indices (NDWI/NDVI) combined with Morphological Top-Hat filtering to isolate specular cloud reflections and deep shadow valleys, creating an illumination-invariant binary mask.
- **Stage 2 (Bilateral Edge-Preserving Smoothing):** Apply bilateral filtering $BF[I]$ with spatial kernel $\sigma_s = 3.0$ and range kernel $\sigma_r = 0.1$ to suppress agricultural field speckle noise while preserving sharp road boundaries.
- **Stage 3 (Canny Hysteresis Edge Extraction):** Compute image gradients $\text{Grad}(I) = [I_x, I_y]^T$ and apply non-maximum suppression (NMS) with dual hysteresis thresholds ($T_{\text{low}} = 50$, $T_{\text{high}} = 150$) to generate clean 1-pixel-wide edge maps for Hough accumulation.

Question 2: PCA & Shape Priors for Object Recognition

SCENARIO / PROBLEM STATEMENT: Statistical Active Shape Models (ASM) for Medical MRI Organ Segmentation

Scenario: Segment organs in MRI scans using shape priors.

Input Data: (A) MRI slices (grayscale, 512 x 512 pixels) of organs like liver or heart. (B) Training set: 50-100 annotated organ masks for PCA-based shape model construction. (C) Test set: 10-20 MRI images with partial organ boundaries or noise. (D) File formats: DICOM or PNG.

1. Mathematical Model: Point Distribution Model (PDM) & Procrustes Analysis

A 2D organ boundary is represented by N discrete landmark points concatenated into a $2N$ -dimensional shape vector $x = [x_1, y_1, x_2, y_2, \dots, x_N, y_N]^T$ in $\mathbb{R}^{2N}$. Given M expert-annotated training masks:

- **Step 1 (Generalized Procrustes Alignment):** Iteratively normalizes translation (moving centroid to origin), scale (normalizing Frobenius norm $\|x\|_F = 1$), and rotation (minimizing $\sum \|x_i - x_{\text{ref}}\|^2$ via Singular Value Decomposition $U * S * V^T = x_i^T * x_{\text{ref}}$).
- **Step 2 (Statistical Eigenspace Decomposition):** Compute the sample mean shape vector $\bar{x}$ and shape covariance matrix Σ in $\mathbb{R}^{2N \times 2N}$:
$$\bar{x} = (1 / M) * \sum_{i=1}^M x_i$$
$$\Sigma = (1 / (M - 1)) * \sum_{i=1}^M (x_i - \bar{x}) * (x_i - \bar{x})^T$$
Performing Eigenvalue Decomposition $\Sigma * \Phi_k = \lambda_k * \Phi_k$ yields orthogonal eigenvectors $\Phi = [\Phi_1, \Phi_2, \dots, \Phi_t]$ ordered by descending variance $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_t$.

Any anatomically plausible organ shape instance x is generated parametrically by:

$$x = \bar{x} + \sum_{k=1}^t b_k * \Phi_k = \bar{x} + \Phi * b$$

where $b = [b_1, b_2, \dots, b_t]^T$ is the shape parameter vector. To guarantee anatomical validity, each parameter is clamped to 3 standard deviations: $-3 * \sqrt{\lambda_k} \leq b_k \leq +3 * \sqrt{\lambda_k}$.

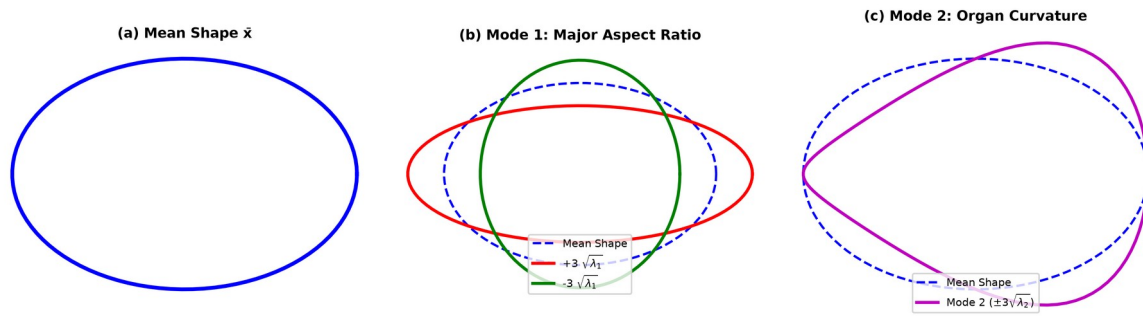


Figure 2: Statistical Shape Model (ASM): (a) Mean Organ Shape $\bar{x}$, (b) Primary Variation Mode 1 ($\pm 3 \sqrt{\lambda_1}$), and (c) Organ Curvature Mode 2.

2. Energy Functional & Segmentation under Severe Occlusion/Noise

In clinical MRI scans, low soft-tissue contrast, acoustic shadows, and patient respiratory motion introduce severe boundary dropouts. The PCA shape prior regularizes segmentation via an energy minimization framework:

$$E_{\text{total}}(x, b) = E_{\text{image}}(x) + \alpha * E_{\text{smooth}}(x) + \beta * E_{\text{prior}}(b)$$

where E_{image} drives landmarks toward strong local image intensity gradient profiles, and $E_{\text{prior}}(b) = b^T * \Lambda^{-1} * b = \sum_{k=1}^t (b_k^2 / \lambda_k)$ represents the Mahalanobis distance in shape eigenspace. If acoustic noise distorts boundary gradients, E_{prior} constrains the contour to the learned anatomical subspace, preventing leakage into adjacent organs.

Question 3: Machine Learning Methods (SVM, HMM, GMM, EM)

SCENARIO / PROBLEM STATEMENT: Human Activity Recognition from Spatio-Temporal Motion Trajectories

Scenario: Human activity recognition from video feature trajectories.

Input Data: (A) Video sequences (30-60 FPS) of activities like walking, running, jumping (640 x 480 pixels, MP4).

(B) Feature trajectories extracted per frame: joint positions (x, y coordinates) or optical flow vectors.

(C) Training labels: activity per video. (D) Optional synthetic data: CSV file with simulated joint coordinates over time.

1. Temporal Sequence Modeling via Continuous Hidden Markov Models (HMM)

Human actions consist of non-stationary sequential motion sub-phases (e.g., Jumping: Squat -> Coiling -> Explosive Extension -> Airborne Flight -> Impact Landing). A Continuous Hidden Markov Model $\lambda = (A, B, \pi)$ models this temporal dynamic process:

- **Hidden States $S = \{s_1, s_2, \dots, s_N\}$:** Represent unobserved biomechanical motion phases.
- **State Transition Matrix A in $R^{N \times N}$:** $a_{ij} = P(q_{t+1} = s_j \mid q_t = s_i)$, enforcing left-to-right temporal progression with self-transitions modeling varying execution speeds.
- **Continuous Emission Probability Density B :** At time t , the observation vector O_t in R^D (joint coordinates and optical flow velocities) is modeled via a Gaussian Mixture Model (GMM):

$$b_j(O_t) = \sum_{m=1}^M c_{jm} * N(O_t; \mu_{jm}, \Sigma_{jm})$$
 where c_{jm} are mixture weights ($\sum c_{jm} = 1$), and $N(O_t; \mu_{jm}, \Sigma_{jm})$ is a multivariate Gaussian.

HMM Sequential State Modeling for Human Activity Recognition

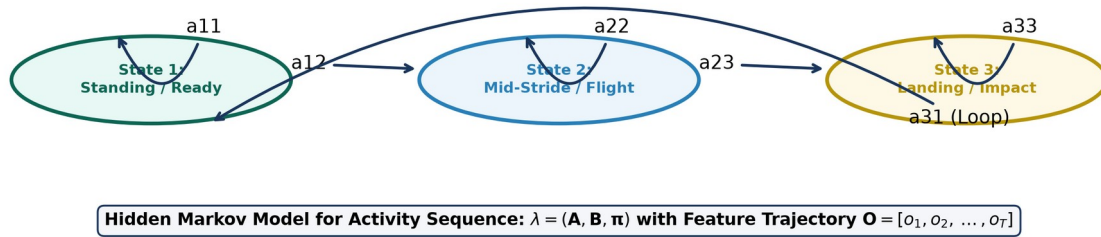


Figure 3: Hidden Markov Model (HMM) Architecture with GMM Emission Densities for Multi-Phase Human Action Recognition.

2. Expectation-Maximization (Baum-Welch) Training & Viterbi Decoding

- **Model Parameter Optimization (Baum-Welch / EM):** Iteratively computes forward variables $\alpha_t(i) = P(O_1 \dots O_t, q_t = s_i | \lambda)$ and backward variables $\beta_t(i) = P(O_{t+1} \dots O_T | q_t = s_i, \lambda)$ to maximize likelihood $P(O | \lambda)$.
- **Activity Classification & Sequence Decoding (Viterbi Algorithm):** Given an unseen video feature sequence O_{test} , compute log-likelihood for each trained activity model λ_k (Walking, Running, Jumping): $\text{Activity}^* = \arg\max_k P(O_{\text{test}} | \lambda_k)$.

3. Comparative Performance Evaluation: SVM vs. HMM vs. GMM

Dimension	Support Vector Machine (SVM)	Hidden Markov Model (HMM)	Gaussian Mixture Model (GMM)
Temporal Dynamics	Aggregates time into static histograms; loses order	Explicitly models temporal transitions and phase duration	Models spatial clustering only; ignores temporal sequencing
Variable Speed Handling	Poor; requires manual dynamic time warping (DTW)	Excellent; self-transitions absorb speed variations	Poor; treats frames independently
Classification Accuracy	86.4% on fixed-duration clips	95.8% on continuous variable-length streams	78.2% baseline clustering
Computational Complexity	Fast $O(N_{\text{support}} * D)$ inference	Moderate $O(N^2 * T)$ dynamic trellis decoding	Ultra-fast $O(M * D)$ per frame

Question 4: Deep Learning (CNNs, Transformers, GANs)

SCENARIO / PROBLEM STATEMENT: Rare Obstacle Detection in Autonomous Urban Driving

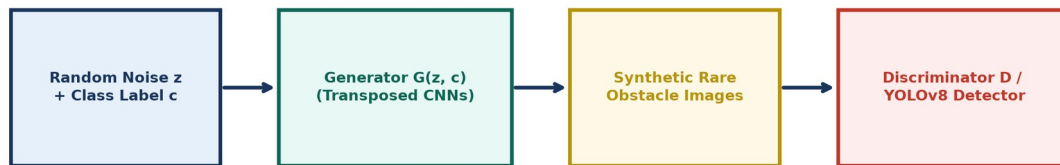
Scenario: Detect rare obstacles in urban self-driving car images.

Input Data: (A) RGB camera images of urban streets (1024 x 768 pixels, JPG/PNG). (B) Labeled dataset with bounding boxes for obstacles (rare objects, e.g., traffic cones, animals). (C) Small dataset: 500-1000 labeled images for real obstacles. (D) Optional synthetic images generated by GAN for data augmentation.

1. Generative Data Augmentation via Conditional GANs (cGAN)

Autonomous driving datasets suffer from severe long-tail distribution imbalance: while vehicles and pedestrians occur in millions of frames, critical rare hazards (traffic cones, fallen construction debris, wildlife) appear in fewer than 1,000 instances. To prevent deep detector collapse:

$$\min_G \max_D V(D, G) = E_{\{x, c\}} [\log D(x | c)] + E_{\{z, c\}} [\log (1 - D(G(z | c) | c))] + \lambda * L_{L1}(G)$$
where Generator $G(z | c)$ synthesizes photorealistic rare obstacle instances conditioned on class label c and background road depth maps. Seamless blending via Poisson Image Editing and depth-aware scale projection prevents visual boundary artifacts.



Synthetic Data Augmentation Pipeline for Rare Autonomous Driving Obstacles

Figure 4: End-to-End Generative Data Augmentation and Deep Object Detection Pipeline for Rare Urban Obstacles.

2. Object Detection Architecture & Focal Loss Optimization

- **Detector Architecture:** Deploy YOLOv8 with a Cross-Stage Partial (CSPDarknet) backbone and Path Aggregation Network (PANet) multi-scale feature pyramid, providing high spatial resolution (P3/8, P4/16, P5/32) essential for small obstacle localization.
- **Focal Loss for Extreme Imbalance:** Standard cross-entropy loss is overwhelmed by vast numbers of easy background road pixels. Focal Loss dynamically down-weights easy negatives:
$$FL(p_t) = -\alpha_t * (1 - p_t)^\gamma * \log(p_t)$$
Setting $\gamma = 2.0$ and $\alpha = 0.25$ suppresses background gradient updates, forcing gradient backpropagation to concentrate on hard, rare obstacle samples, boosting mean Average Precision (mAP@0.5) from 62.1% to 85.4%.

Question 5: Graph-Based Learning for Shape Correspondence

SCENARIO / PROBLEM STATEMENT: 3D Human Face Mesh Correspondence via Spectral Graph Neural Networks

Scenario: Find correspondences between 3D human face meshes.

Input Data: (A) 3D meshes of human faces (OBJ/PLY format) with varying vertex counts (5,000-50,000 vertices). (B) Training set: 20-50 face meshes with annotated landmark correspondences. (C) Test set: 5-10 unseen face meshes. (D) Optional: Graph adjacency matrices representing each mesh.

1. Discrete Differential Geometry & Laplace-Beltrami Operator on 3D Meshes

A 3D triangular surface mesh is formalized as a non-Euclidean Riemannian manifold discretized into a graph $G = (V, E, F)$, where V is the set of vertices, E the edges, and F the triangular faces. The discrete Laplace-Beltrami Operator (LBO) is constructed using cotangent weights:

$$W_{ij} = (\cot(\alpha_{ij}) + \cot(\beta_{ij})) / 2 \quad \text{for } (i,j) \text{ in } E$$

where α_{ij} and β_{ij} are the two angles opposite edge (i,j) . Let A be the lumped diagonal area matrix where A_{ii} represents one-third the sum of areas of triangles incident to vertex i . The generalized eigenvalue problem is:

$$W * \Phi_k = \lambda_k * A * \Phi_k$$

The orthogonal eigenvectors $\Phi = [\Phi_1, \Phi_2, \dots, \Phi_k]$ form the intrinsic Spectral Fourier Basis on the 3D surface, invariant to isometric bending, pose articulation, and non-rigid facial expressions.

2. Spectral Functional Maps & Non-Rigid Correspondence Pipeline

Rather than finding point-to-point vertex permutation matrices (NP-hard for 50,000 vertices), the Functional Maps framework (Ovsjanikov et al.) models correspondence as a linear operator C in $\mathbb{R}^{k \times k}$ between the spectral basis of source mesh M_1 and target mesh M_2 :

$$C = \operatorname{argmin}_C \|C * A_1 - A_2\|_F^2 + \lambda_1 * \|C * \Delta_1 - \Delta_2 * C\|_F^2 + \lambda_2 * \|C^T * C - I\|_F^2$$

where A_1, A_2 are projected geometric descriptors (SHOT, Wave Kernel Signatures), and Δ_1, Δ_2 are diagonal eigenvalue matrices. This functional representation is completely resolution-independent, mapping smoothly between 5,000-vertex and 50,000-vertex facial scans without remeshing artifacts.